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Theoretical study of graded bandgap CZTSSe solar cells with two absorber layers

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Abstract
In this work, new $Cu_2ZnSn(S_{1-x}, Se_x)_4$ (CZTSSe) solar cells consisting of two absorber layers with constant and graded bandgaps are proposed. CZTS and CZTSe are used as the constant bandgap absorbing layers of the first and second proposed cells, respectively. To validate the simulations, first, a graded bandgap CZTSSe-based cell containing AZO/ZnO/CdS/CZTSSe/$MoSe_2$/Mo layers as the basic structure is investigated, revealing that the results are in good agreement with previous experimental data. The impact of varying acceptor doping concentration and temperature on the performance of proposed cells is investigated. The optimum thicknesses of the absorber layers are obtained in all cases. After optimization, maximum efficiencies of 20.99% and 21.37% are obtained for the first and second cells respectively which are significantly greater than the basic structure. The proposed structures provide higher open circuit voltage and short circuit current, compared to the basic structure.

Keywords Bandgap grading · CZTSSe · SCAPS-1D · Thin-film solar cell

1 Introduction

Thin-film solar cells have good potential for application in low-cost photovoltaic devices. Quaternary kesterite-type mineral semiconductors are promising candidates for achieving low-cost and eco-friendly thin-film solar cells (Amiri and Dehghani 2020; Gershon et al. 2015; Patel and Ray 2014; Wanda et al. 2016). Recently $Cu_2ZnSn(S, Se)_4$ as a non-toxic absorber layer with abundant elements, is proposed as a substitution for CIGS layer (Amiri and Dehghani 2020; Cao et al. 2012; Cherouana and Labbani 2017; Hwang et al. 2017). CZTSSe has tunable bandgap energy in the range of 1–1.5 eV and its absorption coefficient exceeds 10^4 cm^{-1} (Cao et al. 2012; Cherouana and Labbani 2017; Kim et al. 2014). The maximum efficiency reported for constant bandgap CZTS solar cells is 11.01% (Yan et al. 2018). Using a graded bandgap structure for CZTSSe has increased efficiency to 12.6% (Gupta and Dixit 2018; Hwang et al. 2017) because recombination near back contact decreases in this case. So, compared to a CIGS solar cell with an efficiency of 21.7%,

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efficiency of CZTSSe cells must be further improved (Amiri and Dehghani 2020; Gupta and Dixit 2018; Hwang et al. 2017). For this reason, it is important to increase both short circuit current and open circuit voltage parameters.

If the bandgap of the absorber layer increases near its back-contact side, recombination rate at the back-contact decreases and collection of generated carriers is enhanced due to the additional induced drift field (Adewoyin et al. 2017; Kato et al. 2014; Ferhati and Djeffal 2018; Mohammadnejad and Prashkouh 2017; Simya et al. 2016). On the other hand, the open circuit voltage (V_{oc}) improves by the increased bandgap in the space charge region.

In this work, new structures for $Cu_2ZnSn(S_{1-x}, Se_x)_4$ (CZTSSe) solar cells using graded and constant bandgap absorber layers are proposed. First, a graded bandgap CZTSSe-based cell containing AZO/ZnO/CdS/CZTSSe/$MoSe_2$/Mo layers as the basic structure which has been experimentally investigated earlier was examined to confirm the validity of simulations. In the first proposed structure, a CZTS absorber layer with constant bandgap is applied between CZTSSe and $MoSe_2$ layers. In the second structure, a CZTSe absorber layer with constant bandgap was placed between the CdS and CZTSSe layers. The effect of temperature and acceptor doping of constant bandgap absorber layer on the performance of cells is also investigated.

2 Simulation method

Simulation of the cells was carried out by using solar cell capacitance simulator (SCAPS-1D) software that is a one-dimensional program designed to examine the electrical characteristics of thin-film solar cells at different operating temperatures (Burgelman et al. 2018; Mohammadnejad and Prashkouh 2017; Simya et al. 2016).

For the simulation of solar cells, Poisson and continuity equations should be solved self-consistently (Burgelman et al. 2018; Mohammadnejad and Prashkouh 2017; Simya et al. 2016). For this reason, the carrier generation rate is used as the input of continuity equations. Charge and potential are calculated at each point and current is obtained by substituting them into the drift–diffusion equations (Burgelman et al. 2018; Mohammadnejad and Prashkouh 2017; Simya et al. 2016).

Bandgap energy of CZT (S_{1-x}, Se_x) depends on the molar ratio of Se to S + Se, x and can be written as (Mohammadnejad and Prashkouh 2017; Song et al. 2014)

$$E_g(x) = (1-x)E_g(CZTS) + xE_g(CZTSe) - 0.07x(1-x) \tag{1}$$

By increasing the mole fraction of sulfur, the bandgap would increase. By changing x from 1 to 0, the Bandgap of CZTSSe compound varies from 1 eV (for CZTSe) to 1.5 eV (for CZTS) (Amiri and Dehghani 2020; Mohammadnejad and Prashkouh 2017). The linear grading model in SCAPS was used to simulate the graded bandgap absorber layer. Defining layers with graded parameters in the form of $A_{1-y}B_y$ is an important ability of SCAPS. The properties of compound materials varying between A and B along layer thickness are defined as a function of y (Burgelman et al. 2018; Mohammadnejad and Prashkouh 2017; Simya et al. 2016)

$$p = f(y) \tag{2}$$

For linear grading, p can be written as (Burgelman et al. 2018; Mohammadnejad and Prashkouh 2017):

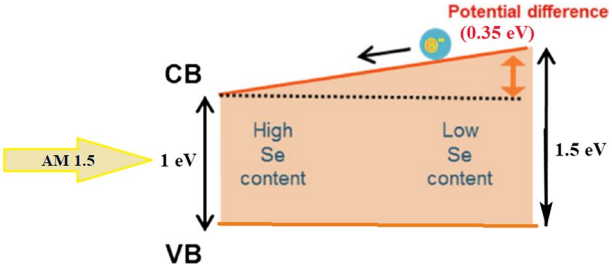


Fig. 1 Schematic diagram of the band alignment in the graded bandgap CZTSSe layer (Woo et al. 2013)



Fig. 2 The structure of basic graded bandgap CZTSSe solar cell

$$p(y) = \frac{p_A(y_B - y) + p_B(y - y_A)}{y_B - y_A} \tag{3}$$

where y_A and y_B are composition values in A and B, respectively. According to Fig. 1, by increasing the bandgap of CZTSSe, electron affinity (χ) decreases. Based on the experimental data (Woo et al. 2013), the electron affinity of CZT (S_{1-x}, Se_x) can be written as (Hironiwa et al. 2014; Minbashi et al. 2017; Woo et al. 2013)

$$\chi = 4.505 - 0.35x \tag{4}$$

where $x = \mathrm{Se}/(\mathrm{S} + \mathrm{Se})$.

In all simulations, the impact of defects and common recombination mechanisms such as Shockley–Read–Hall (SRH) and radiative recombination, are considered. Simulations are performed at 300 K under AM1.5 radiation.

3 Basic cell

The structure of the basic cell illustrated in Fig. 2, is defined according to previous experimental work (Yang et al. 2016) containing AZO/ZnO/CdS/CZTSSe/$MoSe_2$/Mo layers with the respective thicknesses of 300 nm, 50 nm, 50 nm, 1.8 µm, 50 nm, and 600 nm (Yang et al. 2016). Here, aluminum-doped zinc oxide (AZO), intrinsic zinc oxide (ZnO), cadmium sulfide (CdS), copper zinc tin sulfur selenide (CZTSSe), molybdenum selenide ($MoSe_2$) and molybdenum (Mo) layers are used as the transparent electrode, window,

buffer, absorber, and interfacial layers, and back contact, respectively. Based on previous experimental reports, a thin $MoSe_2$ layer is formed between CZTSSe and Mo back contact (Wang et al. 2014; Yang et al. 2016), preventing ohmic contact at the junction of absorber layer and back contact. Therefore, we considered this interfacial layer in the simulations to increase accuracy and validity. Moreover, the experimentally reported series resistance of 0.96 Ω was considered as another important parameter, to have a reasonable analysis (Yang et al. 2016).

Table 1 presents the physical parameters of the materials used in simulations (Hironiwa et al. 2014; Minbashi et al. 2017; Yang et al. 2016; Wang et al. 2014; Woo et al. 2013). The absorption coefficient of $MoSe_2$, CdS, ZnO, and AZO are extracted from data files of the SCAPS program (Burgelman et al. 2018). As shown in Fig. 3, the simulation results are in good agreement with the experimental data (Yang et al. 2016).

Shorter wavelengths of sunlight have more intensity, so more photons can be absorbed in shorter wavelengths. By increasing the bandgap energy of CZTSSe, it can absorb more light due to the higher compatibility with the sun spectrum (Ferhati and Djeffal 2018; Mohammadnejad and Prashkouh 2017; Simya et al. 2016).

In general, bilayer absorber solar cells show a better performance than the cells including a monolayer absorber. In this paper, bilayer absorber cells containing constant (CZTS or CZTSe) and graded bandgap (CZTSSe) layers are investigated. The gradually increased bandgap toward the back contact establishes additional drift field for minority electrons in the p-type absorber layer (Ferhati and Djeffal 2018; Mohammadnejad and Prashkouh 2017; Simya et al. 2016). In other words, the drift field causes the better collection of carriers, so recombination near the back contact is reduced (Ferhati and Djeffal 2018; Mohammadnejad and Prashkouh 2017; Simya et al. 2016).

Table 1 Physical parameters of materials used in simulations (Hironiwa et al. 2014; Minbashi et al. 2017; Yang et al. 2016; Wang et al. 2014; Woo et al. 2013)

Material properties	CZTS	CZTSe	CZTSSe	MoSe2	CdS	ZnO	AZO
Bandgap (eV)	1.5	1	Variable	1.06	2.4	3.4	3.4
Electron affinity (eV)	4.155	4.505	Variable	4.372	4.45	4.6	4.6
Dielectric permittivity	13.6	13.6	13.6	13.6	10	9	9
Density of states in CB (cm^{-3})	2.2×10^{18}	2.2×10^{18}	2.2×10^{18}	2.2×10^{18}	1×10^{18}	1×10^{18}	1×10^{18}
Density of states in VB (cm^{-3})	1.8×10^{19}	1.8×10^{19}	1.8×10^{19}	1.8×10^{19}	1×10^{19}	1×10^{19}	1×10^{19}
Electron mobility (cm^2/Vs)	100	100	100	100	100	100	100
Hole mobility (cm^2/Vs)	25	25	25	25	25	25	25
Donor concentration (cm^{-3})	0	0	0	0	1×10^{18}	1×10^{17}	1×10^{18}
Acceptor concentration (cm^{-3})	1×10^{15}	1×10^{15}	1×10^{15}	1×10^{16}	0	0	0
Defect type	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
Defect density (cm^{-3})	10^{15}	10^{15}	3×10^{14}	10^{14}	10^{16}	10^{15}	10^{15}
Electron capture cross section (cm^2)	10^{-14}	10^{-14}	10^{-14}	10^{-14}	10^{-16}	10^{-16}	10^{-16}
Hole capture cross section (cm^2)	10^{-16}	10^{-16}	10^{-16}	10^{-16}	10^{-14}	10^{-14}	10^{-14}
Absorption coefficient (cm^{-1})	2×10^{4}	2×10^{4}	2×10^{4}	1×10^{1}	2×10^{5}	1.7×10^{5}	1.7×10^{5}

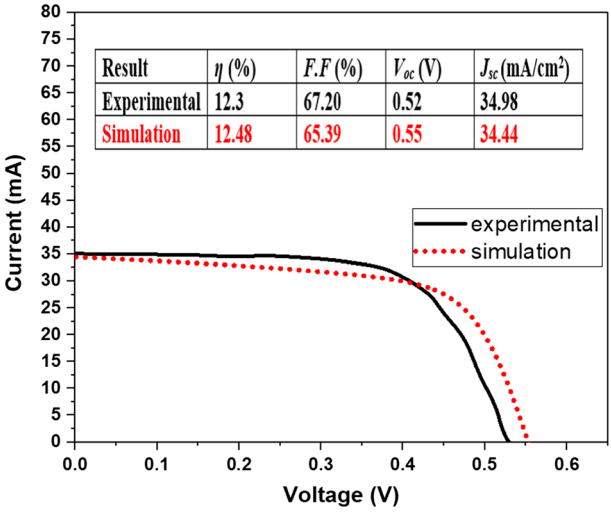


Fig. 3 Comparison of experimental results (Yang et al. 2016) with simulation results obtained for basic CZTSSe cell

4 Bilayer absorber CZTSSe solar cells

4.1 Cell including graded bandgap CZTSSe and constant bandgap CZTS

Due to the advantages of bilayer absorbers, we applied a thin CZTS absorber layer with constant bandgap between CZTSSe and $MoSe_2$ layers. Figure 4 shows the structure of the proposed bilayer absorber cell. By increasing the bandgap of CZTSSe, compatibility with the solar spectrum is enhanced. However, the enhancement of bandgap increases the number of recombination centers that are not desired. By adding a constant bandgap absorber layer, recombination would decrease because an electric field is formed at the interface which introduces a barrier to minority carrier flow to the rear surface.

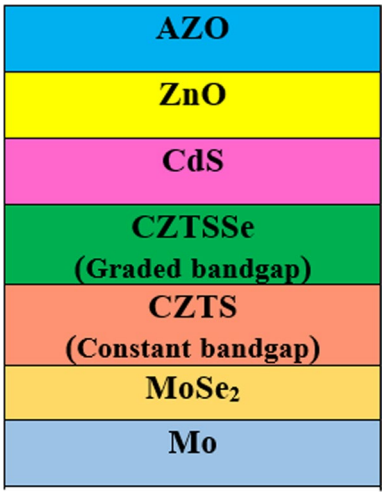


Fig. 4 Structure of bilayer cell including graded bandgap CZTSSe and constant bandgap CZTS absorbing layers

Therefore, electrons cannot reach the back contact easily and recombination is reduced, thus most of them would reach the front contact and the cell performance is improved.

The impact of absorber layers thickness on the performance of this cell is shown in Fig. 5. Typically, by increasing the thickness of the absorber layer to an optimum value optical absorption and efficiency are increased. However, above this optimum thickness efficiency starts to decrease (or saturate), due to the enhancement of recombination rate which is a result of thickness increment. In this cell, the optimum thicknesses of 2.8 µm and 1 µm were obtained for CZTSSe and CZTS layers respectively, leading to electrical parameters of; $\eta = 20.26\%$, $V_{oc} = 0.70$ V, $J_{sc} = 40.15$ mA/cm^2, and *F.* $F = 71.89\%$.

4.1.1 Effect of CZTS acceptor concentration and temperature

CZTS and CZTSSe are intrinsically p-type materials, whose carrier concentration depends on the structural defects and stoichiometry (Romanyuk et al. 2019). Their conductivity is a function of the population of acceptor-like defects such as Cu vacancies (V_{Cu}) and Cu_{Zn} anti-sites which depend on Cu/(Zn + Sn) and Zn/Sn ratios. Alkali elements such as Na are also effective on the doping concentration of the kesterite materials (Romanyuk et al. 2019). It has been shown that by changing the Sn content in CZTS the intrinsic doping is varied over a wide range of 10^{14}–10^{18} cm^{-3} (Romanyuk et al. 2019; Haass et al. 2018).

Light elements such as Na have been used for external doping of CZTS. For this reason, the CZTS layer is immersed in the prepared alkali solution before annealing (Altamura et al. 2016). The maximum value of CZTS hole concentration reported in the earlier experiments is 3.1×10^{20} cm^{-3} (Chen et al. 2013).

Here, the impact of CZTS acceptor concentration in the range of 10^{14}–10^{19} cm^{-3} on the performance of the cell is investigated as shown in Fig. 6. By increasing the acceptor concentration of the CZTS layer, depletion width is reduced and the electric field is increased (Meher et al. 2016; Vallisree et al. 2018). Furthermore, the recombination of carriers decreases and V_{oc} increases which results in efficiency increment. The maximum efficiency of 20.99% was achieved in the acceptor concentration of 10^{18} cm^{-3}, after that efficiency saturates because absorption and recombination rate will be equal at the neutral region (Meher et al. 2016; Vallisree et al. 2018).

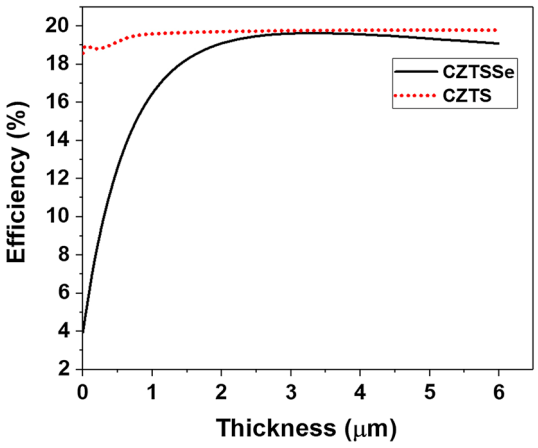


Fig. 5 The efficiency of CZTSSe/CZTS bilayer absorber cell as a function of absorber layers thickness

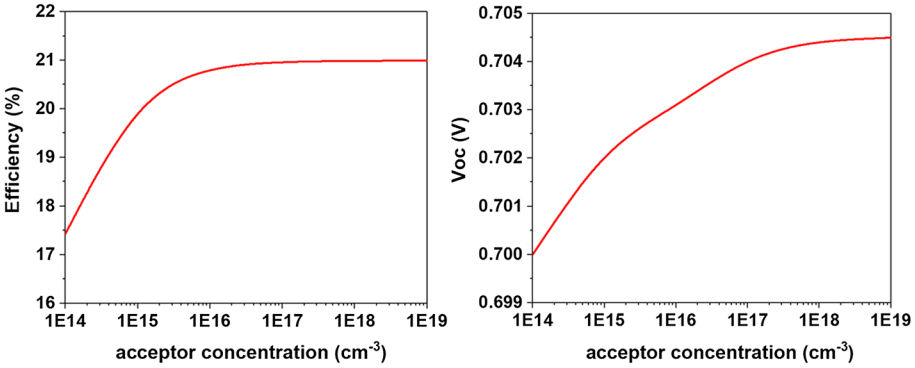


Fig. 6 Efficiency and open circuit voltage of CZTSSe/CZTS bilayer absorber cell as a function of CZTS acceptor concentration

The increment of open circuit voltage by increasing acceptor concentration (N_A) can also be justified using the following relation (Simya et al. 2016)

$$V_{oc} = \frac{E_g}{q} - \frac{KT}{q} \ln\left(\frac{qD_e N_C N_V}{J_{sc} N_A L_e}\right) \tag{5}$$

where E_g is the bandgap energy, q is the electron charge, K is the Boltzmann constant, T is the temperature in Kelvin, D_e and D_h are diffusion constants of electron and hole, N_C and N_V are the conduction band and valence band density of states, J_{sc} is the short circuit current, N_A is the acceptor concentration, and L_e is the diffusion length of electrons (Simya et al. 2016).

The operating temperature of CZTSSe solar cells may vary depending on the season and geographical area. The impact of temperature variations in the range of −30 to 60 °C on the performance of the cell is thus investigated. By increasing temperature, the recombination of carriers increases while V_{oc} decreases, leading to efficiency reduction from 22.48 to 18.1%, as shown in Fig. 7 (Amiri and Dehghani 2020; Jhuma et al. 2019).

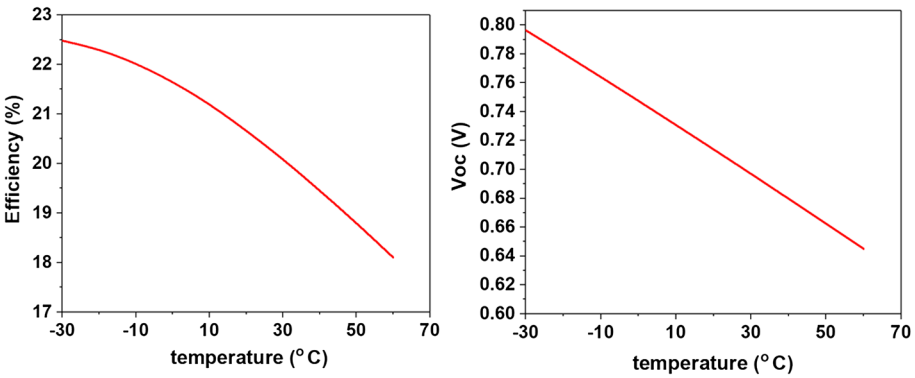


Fig. 7 Efficiency and open circuit voltage of CZTSSe/CZTS bilayer absorber cell as a function of temperature

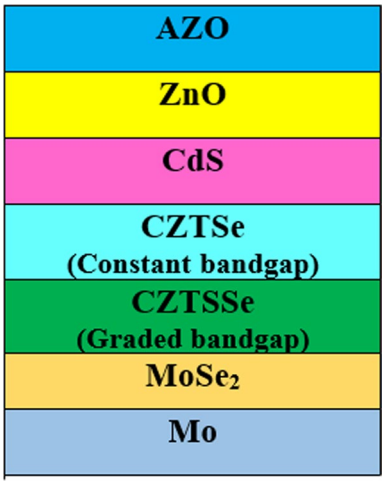


Fig. 8 Bilayer cell including graded bandgap CZTSSe and constant bandgap CZTSe

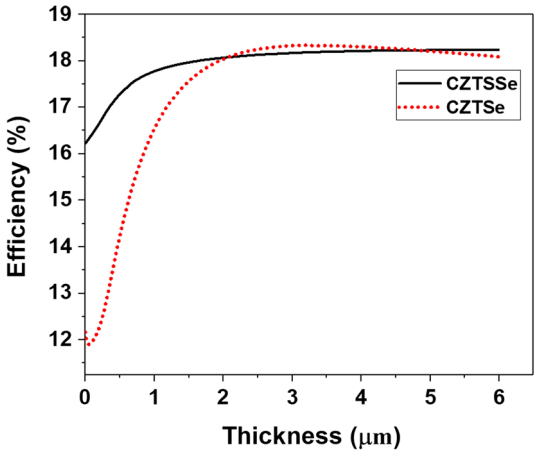


Fig. 9 The efficiency of CZTSSe/CZTSe bilayer absorber cell as a function of absorber layers thickness

4.2 Cell including graded bandgap CZTSSe and constant bandgap CZTSe

In the second proposed structure, as shown in Fig. 8, a CZTSe absorber layer with constant bandgap was applied between CdS and CZTSSe layers. Figure 9 shows the effect of the absorber layers thickness on the efficiency of this cell. The obtained optimum thicknesses of CZTSSe and CZTSe absorber layers are respectively 1.8 μm and 2.5 μm for this cell. The electrical parameters of this cell are; $\eta = 18.24\%$, $V_{oc} = 0.56$ V, $J_{sc} = 45.08$ mA/cm^2, and *F. F* = 71.04%.

4.2.1 Effect of CZTSe acceptor concentration and temperature

The impact of acceptor concentration of the CZTSe layer in the range of 10^{14}–10^{19} cm^{-3} on the performance of the solar cell is investigated. According to Eq. (5), by increasing the acceptor concentration of CZTSe, V_{oc} increases and so efficiency grows up, as shown in

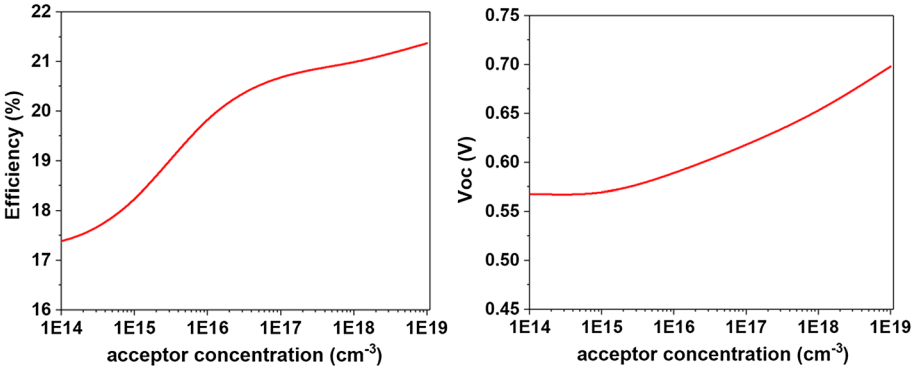


Fig. 10 Efficiency and open circuit voltage of CZTSSe/CZTSe bilayer absorber cell as a function of CZTSe acceptor concentration

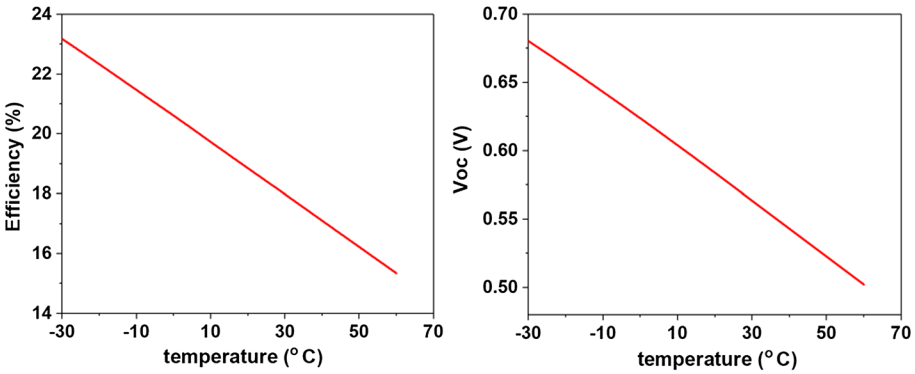


Fig. 11 Efficiency and open circuit voltage of CZTSSe/CZTSe bilayer absorber cell as a function of temperature

Fig. 10. Maximum efficiency of 21.37% was obtained in the acceptor concentration of 10^{19} cm^{-3}.

The impact of temperature variations on the electrical parameters of cell was also investigated. As shown in Fig. 11, by increasing temperature from −30 to 60 °C the efficiency decreases from 23.18 to 15.34% which is a result of reduction in open circuit voltage.

The effect of temperature on the efficiency of proposed cells was shown in Fig. 7 and Fig. 11, revealing that the ratio of maximum to the minimum efficiency of the first proposed cell and the second one is 1.24 and 1.51, respectively. Therefore, the variation of efficiency with temperature is significantly higher in the second structure. So, the first structure is more stable and the environmental conditions have a smaller effect on its performance.

Simulation results of the two proposed bilayer CZTSSe solar cells at optimum absorber layer thickness and optimum acceptor doping concentration are briefly presented in Table 2, compared to previous works. It is clear that the efficiencies of our bilayer absorber cells are significantly greater compared to monolayer absorber cells either with graded or constant bandgap. Because employing the constant bandgap absorber layer near the graded bandgap absorber layer reduces the recombination. It is also found that the cell including

Table 2 Photovoltaic parameters of two proposed bilayer CZTSSe solar cells, compared to previous works

Absorber layers	η (%)	V_{oc} (V)	J_{sc} (mA/cm^2)	F.F (%)	Thickness of the graded bandgap absorber layer (μm)	Thickness of the constant bandgap absorber layer (μm)
CZTSSe/CZTS (this work)	20.99	0.70	40.18	74.14	2.8	1
CZTSe/CZTSSe (this work)	21.37	0.69	38.49	79.50	1.8	2.5
CZTS (Yan et al. 2018)	11.01	0.73	21.74	69.27	–	1.2
CZTS (Adewoyin et al. 2017)	12.26	1.51	16.41	74.40	3	–
CZTS (Ferhati and Djeffal 2018)	16.9	0.88	28.50	66	1.15	–
CZTSSe (Woo et al. 2013)	7.17	0.50	24.55	57.80	0.78	–
CZTSSe (Yang et al. 2016)	12.3	0.521	34.98	67.20	1.8	–
CZTSSe (Mohammadnejad and Prashkouh 2017)	15.60	0.67	38.81	60.48	2.5	–

CZTSe/CZTSSe absorber layers offers the best performance with maximum efficiency of 21.37%.

5 Conclusion

New bilayer absorber solar cells based on graded bandgap CZTSSe were introduced. In the first structure by applying a 1 μm constant bandgap CZTS layer below a 2.8 μm graded bandgap CZTSSe layer, maximum efficiency of 20.99% was obtained. In comparison, the achieved efficiency for the second structure including a 2.5 μm constant bandgap CZTSe layer on the top of a 1.8 μm graded bandgap CZTSSe layer was as high as 21.37%. In both structures, the acceptor doping concentration and thickness of absorber layers were optimized. It is found that the efficiencies of our bilayer absorber cells are significantly greater than monolayer absorber cells with graded or constant bandgap. The results also show that the variation of efficiency with temperature is higher in the second structure.

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